# Montgomery (Mario) Tan Yu

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**PROFESSIONAL SUMMARY**

Graduate student with 6+ years of international work experience in public, private and fintech startup sectors
Data scientist with proven track record of solving business problems with strong skills in neural network programming, video/image processing, kernel optimization, reinforcement learning, NLP and building ML data pipelines
Developed 29 ML and DL projects using tools like Pandas, PyTorch, TensorFlow, NumPy, Matplotlib, Scikit Learn
Project manager and storyteller with excellent written and verbal communication skills in English, Chinese, and German
Seeking to transition into AI engineering roles for robotics, deep learning and computer vision

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**EDUCATION**

New York University – Manhattan, NY
Master of Science in Data Science, Industry Concentration *Expected Graduation: May 2027*
(*Computer Vision, Reinforcement Learning, Machine Learning, Probability and Statistics, Deep Learning, Big Data)*

UT Austin Postgraduate Program in Artificial Intelligence and Machine Learning *Completed Mar 2025*
(*Python Foundations, Advanced Machine Learning, Neural Networks, Natural Language Processing, Intro to Computer Vision)*

Fordham University – Bronx, NY *Graduated May 2022*
Bachelor of Arts in Economics, Minor in Computer Science – Dean's List, Omicron Delta Epsilon, Fordham Loyola Scholarship

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**PROFESSIONAL EXPERIENCE**

THE WORLD BANK (Washington DC) *Oct 2022 – July 2025*
*Short-Term Consultant, (Data Scientist)*

- Built Project Costing Tool using logistic regression models to predict and estimate airfare travel, hotel costs and project budgets for IBRD projects, saving the World Bank $9 million in additional costs
- Built linear regression model to predict prices of AirBNB rental rooms for World Bank Staff during World Bank-IMF Annual Meetings in Marrakesh in 2023
- Created a new RPA Bot to automate budget transfer process for $160 million worth of new employees and contracts in FY25
- Built multiple ML models to predict mortality rates and hospital childbirth outcomes among lower-income women for every province in Laos - successfully secured additional $15 million for hospital funds from Laos government in 2022 after mortality rates were deemed too high by United Nations
- Worked with IUCN to classify primate species in Latin America based on conservation status
    - Developed a deep learning model using Llama 3 to determine monkey species and status (least concern, vulnerable, endangered, critically endangered) based on photo images taken in Colombia and Costa Rica

FORDHAM UNIVERSITY CAREER CENTER (New York, NY) *Oct 2021 – June 2022*
*Senior Data Analyst*

- Pulled statistics for quarterly and annual CampusLabs reports while working closely with Sr. Associate Director of Career Services to develop strategies for improvement
- Created weekly Power BI reports for a year-to-year overview of all services offered for 16,000 students and their employment outcomes

ORDER.COM
*Operations and Sales Analytics Intern* *Jan 2020 – June 2020*

- Automated existing workflow processes for seven members on the Sales team to increase client acquisition by 20% during COVID-19 pandemic
- Built ML model using internal Salesforce database to classify 500 customers and potential clients for future product initiatives

COMCAST Technology Solutions | Freewheel (London, UK) *Sept 2019 – Dec 2019*
*Business Intelligence Intern*

- Published all UK and EU data for Freewheel's Video Monetization Report Q32019 and Q42019
- Designed wiki for TV environment and emerging Video-on-Demand usage in Baltics countries, demonstrating high potential for live content expansion (showed 31% higher growth of Roku and other set top boxes in EU)

- Extracted PreRoll and MidRoll ad impression data (SQL, Python, AWS) and ad completion rates for all media networks in the EU and UK

**SANTOS | KNIGHT FRANK** *July 2019 – Aug 2019*
*Research and Consulting Intern*

- Created financial modelling and data analysis for Knight Frank Wealth Report 2019
- Created 3 chart and historical financial analyses per week on prevailing contract prices of over 10 million pesos for pre-selling
- apartment units in the Philippines
- Ran historical stress testing simulations to educate 20 Chinese-based clients about new developments in Philippine stock market

ASIAN INSTITUTE OF MANAGEMENT *July 2019 – Aug 2019*
*Summer 2019 Intern*

- Assisted in designing the 2019 – 2020 general curriculum for 30+ MBA students at Zuellig School of Development
- Created and presented 4 bi-weekly presentations on technology trends in Southeast Asia and government projects in India, Pakistan and Vietnam to 4 World Bank representatives
- Wrote 4 speeches for private sector finance professionals

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**INDEPENDENT PROJECT PORTFOLIO** *See GitHub Profile for more Projects*

**Diabetes Risk Prediction**

- Developed ML models to predict onset of Type 2 Diabetes among women in Tamil Nadu, India circa 2023
  - EDA with Bivariate and Univariate Analysis showed positive correlation between class against insulin levels and BMI
  - Built Random Forest, Decision Tree and Bagging Classifier algorithms to optimize for recall value
  - After hyperparameter tuning, training Random Forest model revealed BMI, Age and Glucose levels were the most significant predictors of Type 2 Diabetes

**Airfare Travel Price Prediction**

- Trained 5 CNN Models to predict Airfare prices for all possible countries and destinations for fiscal year 2024
  - Results were extrapolated to fiscal year 2025 – 2030 due to high recall and F-1 values
  - World Bank HR employees now use this neural network to approach future costing predictions for staff on mission travel
- World Bank employees allowed to reuse and adjust neurons (epoch vs accuracy stats) in the model for future use

**HR Job Change Prediction**

- Built 6 Neural Network models using credentials, demographics, and work experience related data, to predict the probability that World Bank candidates are looking for a new job, and show the main factors affecting employee attrition
  - Developed EDA and Feature Engineering for Bivariate and Univariate Analysis
  - Tested, Trained and evaluated 6 models using ADAM, SGD and Drop normalization
  - Presented Business Insights and Recommendations to World Bank to fund gender diversity initiatives and extensive training for existing employees to increase retention

**Audio MNIST Digit Recognition**

- Bulit Artificial Neural Network using Keras and SkLearn (Librosa package for Python) as an independent project to look at Mel/MFCC spectrograms of audio files and classify them into 10 classes.
  - Visualized Mel Frequency Cepstral Coefficients into a 2D graph using a spectrogram function, showing coefficients against time
  - Used a sequential model with multiple connected hidden layers and an output layer to return a single continuous value
  - The audio files are recordings of different speakers uttering a particular digit and the corresponding class to be predicted is the digit itself
  - ANN achieved 99% recall, precision and F-1 score
  - Test set results were shown to World Bank for further implementation in voice-activated security protocols

**New World Monkey Species Classification for IUCN Endangered Species List**

- Built sequential CNN models (25 epochs, batch size of 64) using TensorFlow library to classify 12 new world monkey species based on pictures I took in Costa Rica and Colombia
  - Applied data augmentation using SGD optimizer to adjust photos for maximum clarity
  - Confusion Matrices and other graphs were utilized by World Bank in further published studies on biodiversity loss in Latin American countries

- Used internal dataset provided by IUCN in collaboration with World Bank (published on Kaggle) to classify these monkey species on the IUCN endangered species list (Least Concern, Vulnerable, Endangered, Critically Endangered)

**Plants Seedling Classification for Evaluation on Risk of Species Extinction**

- Built sequential CNN models (50 epochs, batch size of 64) using TensorFlow library to classify 40 plant seedling species native to Scandinavian countries, based on pictures taken by the Aarhus University Signal Processing Group, in collaboration with the University of Southern Denmark
    - Applied data augmentation using SGD optimizer to adjust photos for maximum clarity
    - Confusion Matrices and other graphs were utilized by World Bank in further published studies on biodiversity loss in Europe-Central Asia countries
- Used internal dataset provided by World Bank (published on Kaggle) to identify plant species based on images and classify these plant seedlings on extinction risk from climate change

**University Admission Prediction for incoming UCLA students**

- Developed a feed-forward neural network using Keras to predict chances of university admission based on real data from UCLA uploaded to Kaggle
    - Trained and evaluated 4 deep learning models using accuracy vs epoch curve on adamax optimizer for binary classification to achieve 96% accuracy on most accurate model
    - Observations that on average, 28% of students would get into UCLA, with GRE and TOEFL increasing relevance of SOP in admission.

**AirBNB Pricing Model Analysis**

- Developed a Linear Regression ML model for an independent analysis of AirBNB rent prices in Marrakesh, Morocco for World Bank-IMF Annual Meetings in October 9 – 15, 2023
    - First project to use R instead of Python
- Predicted AirBNB room prices in advance of World Bank-IMF meetings for all incoming World Bank personnel per Budget department guidelines
- Price Distribution, Cancellation Policies, Price Variation by Bedrooms and Accommodates results from this study were used in our department's Project Costing Tool for future bookings for World Bank staff on mission travel in other countries

**Deutsche Bank Study on Risk of Credit Default among SE Asian Clients**

- Assisted in developing ML models, EDA and Hyperparameter Tuning on evaluation of selected Deutsche Bank client businesses in SE Asia region
    - Based on internal dataset including Age, Sex, Job, Housing, Credit Amount, Savings Account and Risk, we predicted whether 500,000 selected business in ASEAN countries (Philippines, Malaysia, Indonesia, Singapore, Vietnam, Hong Kong) would default or were highly at risk of defaulting
    - First large-scale ML project was submitted to UT Austin Hackathon and won 11$^{th}$ place among 1000 other contestants in my cohort group in the PG Program for AI and Machine Learning
    - Compared 2 models (Decision Tree and XGBoost) using both GridSearch and RandomizedSearch respectively to determine that XGBoost using GridSeachCV was most reliable with highest accuracy, F1 and recall scores at 95%

**Blackline Users Churn Prediction Study**

- Volunteered my services for World Bank budget department to show that Blackline software contractor was reliable
- Performed extensive feature engineering and EDA to show rates of attrition for Blackline proprietary software used for automating existing office processes
    - Showed full demographic analysis of Blackline customers circa 2020 – 2023 (univariate/bivariate)
- Trained 5 ML Models showing why people adopt Blackline software integration services after 1-4 years
    - Adaboost, Gradient Boosting and Random Forest models all showed highest AUC values
    - All 3 models gave around 85% accuracy and 91% recall values after hyperparameter tuning
    - ML models showed World Bank IFC Board that Blackline software was reliable for existing World Bank HR processes